

# Assignment 1

## Number Systems- Assignment

**Q1.** Complete the table with equivalents in Binary, Octal, and Hexa for: 143.75 (decimal), 11011.101 (binary), 453 (octal), 1F.C (hex).

| Decimal | Binary    | Octal | Hexa |
|---------|-----------|-------|------|
| 143.75  |           |       |      |
|         | 11011.101 |       |      |
|         |           | 453   |      |
|         |           |       | 1F.C |

**Q2.** Find equivalents:

a)  $324_5 = ?_{10}$

b)  $32.23_4 = ?_{10}$

c)  $231_{10} = ?_7$

d)  $189_{10} = ?_6$

**Q3.** Find the radix r:

a)  $(312/20)_r = (13.1)_r$

b)  $264_r + 136_r = 433_r$

c)  $221505_8 = 12345_r$

**Q4.** Addition:

a)  $1101_2 + 11011111_2$

b)  $77652_8 + 76543_8$

c)  $578A_{16} + ABDE_{16}$

d)  $234_6 + 315_6$

**Q5.** Binary  $\rightarrow$  Gray code:

a)  $10101_2$

b)  $01110_2$

**Q6.** Gray code  $\rightarrow$  Binary:

a) 01101 (gray)

b) 10101 (gray)